

1st Sit Coursework Question Paper**Spring Semester 2026**

Module Code:	CC5067NI
Module Title:	Smart Data Discovery
Module Leader:	Mr. Alish K.C. (Islington College)

Coursework Type:	Individual
Coursework Weight:	This coursework accounts for 60% of the overall module grades.
Submission Date:	
Coursework is given out:	Week 4
Submission Instructions:	Submit the following to Islington College's MST Portal before the due date: • A report (document) in .pdf format in the MST Portal or through any medium which the Module Leader specifies. • Associated python program into Zip file
Warning:	London Metropolitan University and Islington College take plagiarism very seriously. Offenders will be dealt with sternly.

PLAGIARISM

You are reminded that there exist regulations concerning plagiarism. Extracts from these regulations are printed overleaf. Please sign below to say that you have read and understood these extracts:

Extracts from University Regulations on Cheating, Plagiarism, and Collusion

Section 2.3: *“The following broad types of offence can be identified and are provided as indicative examples*

- (i) *Cheating: including taking unauthorised material into an examination; consulting unauthorised material outside the examination hall during the examination; obtaining an unseen examination paper in advance of the examination; copying from another examinee; using an unauthorised calculator during the examination or storing unauthorised material in the memory of a programmable calculator which is taken into the examination; copying coursework.*
- (ii) *Falsifying data in experimental results.*
- (iii) *Personation, where a substitute takes an examination or test on behalf of the candidate. Both candidate and substitute may be guilty of an offence under these Regulations.*
- (iv) *Bribery or attempted bribery of a person is thought to have some influence on the candidate's assessment.*
- (v) *Collusion to present joint work as the work solely of one individual.*
- (vi) *Plagiarism, where the work or ideas of another are presented as the candidate's own.*
- (vii) *Other conduct calculated to secure an advantage on assessment.*
- (viii) *Assisting in any of the above.*

Some notes on what this means for students:

1. Copying another student's work is an offence, whether from a copy on paper or a computer file, and in whatever form the intellectual property being copied takes, including text, mathematical notation, and computer programs.

2. Taking extracts from published sources *without attribution* is an offence. To quote ideas, sometimes using extracts, is generally to be encouraged. Quoting ideas is achieved by stating an author's argument and attributing it, perhaps by quoting, immediately in the text, his or her name and year of publication, e.g. “ $e = mc^2$ (Einstein 1905)”. A reference section at the end of your work should then list all such references in alphabetical order of authors' surnames. (There are variations on this referencing system which your tutors may prefer you to use.) If you wish to quote a paragraph or so from published work then indent the quotation on both left and right margins, using an italic font where practicable, and introduce the quotation with attribution.

Coursework Assignment

The coursework is an individual assessment weighted 60% of the marks for the module. It's mainly an exercise in using programming skills for data analysis, showcasing your problem-solving abilities and capacity for critical thinking and evaluation. This assignment involves the **Network Traffic Data Analysis**. You are expected to write Python programs and technical report on data understanding, preparation, exploration, and initial analysis.

Data Set Description

The dataset contains various network flow features such as Destination Port, Flow Duration, Packet Counts, Packet Lengths, and other statistical metrics, along with a label indicating whether the traffic is BENIGN or an attack.

The objective of this analysis is to gain a deeper understanding of the network flow characteristics, identify relationships between features, and prepare the data for further machine learning model training for intrusion detection. Your primary task is to prepare the data for further data mining and classification.

Requirements Specifications

1. Data Understanding

- Understand the dataset and provide a brief description about it explaining some important columns. **[10 Marks]**

2. Data Preparation

- Write a python program to **load data** into pandas Data Frame and display last 50 records. **[5 Marks]**
- Write a Python program to create a separate Data Frame containing only the following columns list for further analysis. **[5 marks]**
columns = ['Destination Port', 'Flow Duration', 'Total Fwd Packets', 'Total Backward Packets', 'Total Length of Fwd Packets', 'Total Length of Bwd Packets', 'Fwd Packet Length Mean', 'Bwd Packet Length Mean', 'Flow Bytes/s', 'Flow Packets/s', 'Packet Length Mean', 'Packet Length Std', 'Average Packet Size', 'Active Mean', 'Idle Mean', 'Label']
- Write a Python program to check for NaN (**missing values**) and remove them from the newly created DataFrame. **[5 Marks]**
- Write a python program to check if **duplicated records** exist and find their total number. **[5 Marks]**

- Write a python program to show total number of columns. Also display name of all columns. **[5 Marks]**
- **Rename the values in the Label column as follows:** **[10 Marks]**
 - BENIGN – **Normal Traffic**
 - Infiltration – **Attack**

3. Data Analysis

- Write a Python program to show summary statistics of **sum, mean, standard deviation, skewness, and kurtosis** of any 2 chosen numeric variable. **[5 Marks]**
- Write a Python program to calculate and show **correlation** of all numeric variables. Display top 5 correlated feature. **[5 Marks]**

4. Data Exploration

- Plot a bar chart showing the frequency of each category in the Label column (with values as **Normal Traffic and Attack**). Determine whether the Label, as the dependent variable, is balanced or not in the dataset. Provide an analysis. **[10 Marks]**
- Determine which category in the Label column (with values as **Normal Traffic and Attack**) has the highest average Flow Duration and the highest average Packet Length Mean. Visualize these findings with pie chart. **[10 Marks]**
- Generate boxplot of '**Fwd Packet Length Mean**' grouped by the Label column to visualize the distribution for each attack type and normal traffic. Analyse the differences across these categories. **[10 Marks]**
- Perform a hypothesis test to determine if there is a significant difference in the mean Flow Duration between **Normal Traffic** and **Attack classes**. State your null and alternative hypotheses, choose an appropriate test, perform it, and interpret the results.

[10 Marks]

5. Document Organization

- Technical Report Structure
- Python code with screenshot comments and output.

[5 Marks]

Milestone1

Students must submit the following:

1. **.ipynb File**
 - **Question 1:** Data Understanding
 - **Question 2:** Data Preparation
2. **Report**
 - **Question 5:** Document Organization

Milestone2

Students must submit the following:

1. **.ipynb File**
 - **Question 1:** Data Understanding
 - **Question 2:** Data Preparation
 - **Question 3:** Data Exploration
 - **Question 4:** Data Exploration / Analysis
2. **Final Report**
 - **Question 5:** Document Organization
 - Should include explanations and results for all questions.

End of paper